



Document forgery detection: a comprehensive review

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Abstract

Technological advancements and image processing software have made document fraud more prevalent, aided by the availability of inexpensive scanners and printers that facilitate document alteration. Document forensics addresses this issue through active and passive techniques for detecting forgeries. Active methods, like using extrinsic fingerprints and signatures, help in straightforward document authentication. In contrast, passive methods require more sophisticated verification techniques. This review article explores various strategies for identifying the source printers of digital documents and the authors of scanned handwritten documents. It examines a range of methodologies, including machine learning and deep learning approaches, relevant to document forensics. Additionally, the article reviews research on detecting different types and models of printers using textual detection and noise analysis. It also discusses classification and feature extraction techniques, providing a comprehensive overview of the current state-of-the-art methods and tools in the field of document forensics.

Keywords Document forensics · Printer forensics · Digital document · Printer source identification

1 Introduction

As the utilization of advanced technologies and the Internet grows rapidly, multinational organizations, the banking system, government sectors, and private companies have adopted digital techniques to manage their daily tasks. Digital documents are a prime example. A digital document refers to any scanned or soft copy of a document (Fig. 1). Since these documents are verified digitally, it becomes a challenging task to ascertain their authenticity whether they are original or forgeries. Fraud can occur through various means, such as altering the document (e.g., changing names or other text details), forging images, and signatures, or tampering with holograms or seals. Around ten thousand cases of document forgery are recorded in Korea every year [40]. The study of questioned documents is a critical component of forensics and the criminal system. Fraudsters and other criminals may manipulate papers during their unlawful activity [36]. Document forensics helps detect various types of document-related fraud.

To preserve the integrity of legal documents, academic records, and financial transactions, forgery detection has become more and more important. The likelihood of fraudulent activity increases as society depends more on printed and digital documents, posing serious threats to both the legitimacy of judicial institutions and the stability of the economy. These techniques employ different approaches to identify forgery in documents, which are divided into text, images, signatures, seals, stamps, and holograms. Each part is examined thoroughly using various feature extraction techniques to detect forgeries. The tasks of document forensics, as defined by Sezonov et al. [54], are as follows: (1) to develop document forensics classification based on different criteria; (2) to develop methods for addressing document forgery; and (3) to develop techniques for reconstructing the document's or its owner's informational content. From the moment a digital image is first captured until it is altered, it goes through several phases: image capturing, picture coding, and image editing [48]. Printer inks are also becoming increasingly important for various applications. Analyzing printer inks can help detect forgery in documents. The classification and discrimination of printed inks in suspicious documents whether civil or criminal can provide valuable evidence regarding their authenticity [70].

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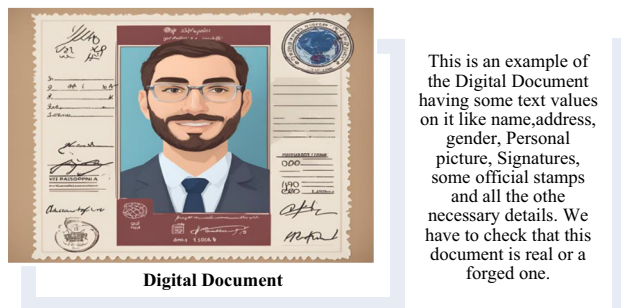


Fig. 1 Digital document

1.1 Source printer identification

This article discusses the techniques used to identify the source of a printer. Identifying the brand and model of a printer plays a significant role in document forensics technology, as it helps detect forgery in documents. Numerous forensic techniques attempt to identify the source device using commercially available recording devices and automated methods. However, finding intrinsic signatures that can uniquely identify the source device and withstand a range of printing process impacts is challenging. Typically, the printer model is the maximum identification rate that can be achieved. Therefore, examining document color tracking dots an external signature present in almost all contemporary color laser printers is crucial [49]. Due to the proliferation of affordable printers in the market, identifying the primary source of a document can be very tedious. Thus, accurately identifying and extracting the unique attributes and features of printers is essential. The feature detection techniques are mostly based on the texture patterns and noise patterns. In the texture patterns, the source printer relies on identifying unique patterns in the document. This approach involves generating numerical identifiers from the textural properties of printed materials to distinguish prints from various printers. Texture or patterns can be detected by examining the spacing between fonts, shapes of the fonts, and font types. In noise-based detection, these patterns are analyzed to classify the printer brand, toner, and model. Noise patterns are influenced by factors such as printing quality, printing style, inconsistent ink distribution, and different types of color noise. Generally, noise patterns in printed documents are affected by a range of mechanical and material elements inherent in the printing process. Devices like cameras can also produce noise due to manufacturing issues, pixel defects, and fixed pattern noise or photo response nonuniformity [5].

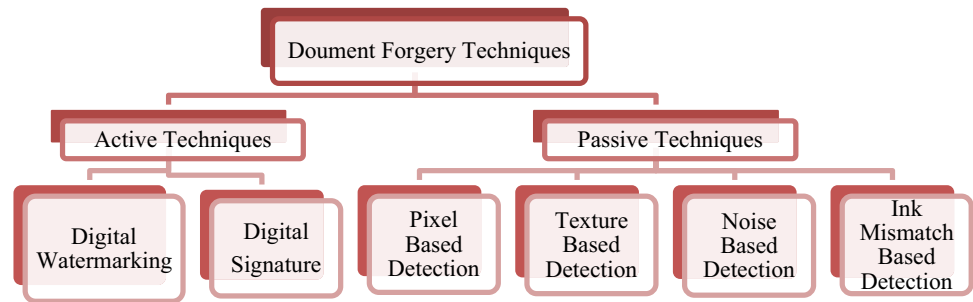
1.2 Classification of document forgery techniques

There are two main types of techniques used in document forensics: active techniques and passive techniques as

depicted in Fig. 2. Active techniques involve data or code embedded in the document before it is transferred, allowing forgery to be detected based on that embedded information.

Various active forensics techniques, such as digital watermarking, hashing, and digital signatures, are employed to verify documents. Digital watermarks protect information about the source and copyright of the document. Many advanced watermarking algorithms achieve high levels of resilience, imperceptibility, and capacity; however, balancing these three conflicting characteristics remains challenging. Khadim et al. [32] developed a new invisible and intelligent method for three-level digital watermarking, enhancing document security during Internet transfers. In visible watermarking, the embedded data or message is visible on the document, whereas invisible watermarking hides the encoded data [58]. Digital signatures offer another way to protect documents using public-key cryptography techniques, including the Rivest–Shamir–Adleman (RSA) public-key encryption algorithm and the digital signature algorithm [59]. Different hash functions are used with the public and private keys of the document owner. Jiang et al. [28] present a comprehensive and adaptable PDF watermarking scheme that can withstand various types of attacks, such as text editing, format modification, page extraction, and textbox republication. Salih et al. [51] developed a method proposing an integrated digital signature technique (SHA-256 + DSA) to provide a digital signature for every document.

In contrast, passive techniques involve examining different parts of the document and extracting various features to detect forgery based on those features. They determine the document's authenticity through statistical distortion analysis. To detect image tampering, passive techniques analyze the raw image using various statistical and semantic aspects of the image content. These techniques rely solely on the statistical features of the image to identify tampering locations. Consequently, algorithms and techniques for image identification and localization based on passive tampering vary according to the security construct employed [56]. Passive techniques examine different essential properties of digital documents and investigate the texture styles of the documents. Different types of printers also introduce unique types of noise on documents. All these properties aid in identifying forgery using passive techniques. Pixel-based methods focus on detecting mathematical alterations or statistical irregularities at the pixel level of the digital document or image [31]. The SVM machine learning method is frequently used for forgery detection due to its accuracy and strong performance on datasets with few samples [6]. Copy-move forgery detection is another major issue in digital document forensics, which involves identifying similar regions in the image or document (Thajeel et al. [60]). Copy-move forgery, also known as cloning, is a frequent technique employed by manipulators to fake digital pictures; hence copy-move

Fig. 2 Classification of document forgery techniques**Fig. 3** Applications of document forensics

forgery detection techniques are among the most essential types of passive detection methods [46]. Despite much study over the previous decade, passive forgery detection remains an open research topic.

1.3 Applications of document forensics

Document forensics can be applied in various areas to check documents and detect forgery. Some of its key applications include (Fig. 3):

- **Fraud detection:** This is one of the most significant applications of document forensics. It is particularly useful for identifying forged signatures and altered text in legal documents, educational records, or other digital documents. Fraud detection can be achieved through various techniques such as handwriting analysis, texture analysis of fonts, and others.
- **Criminal investigations:** Document forensics plays a crucial role in criminal investigations. It is used to examine

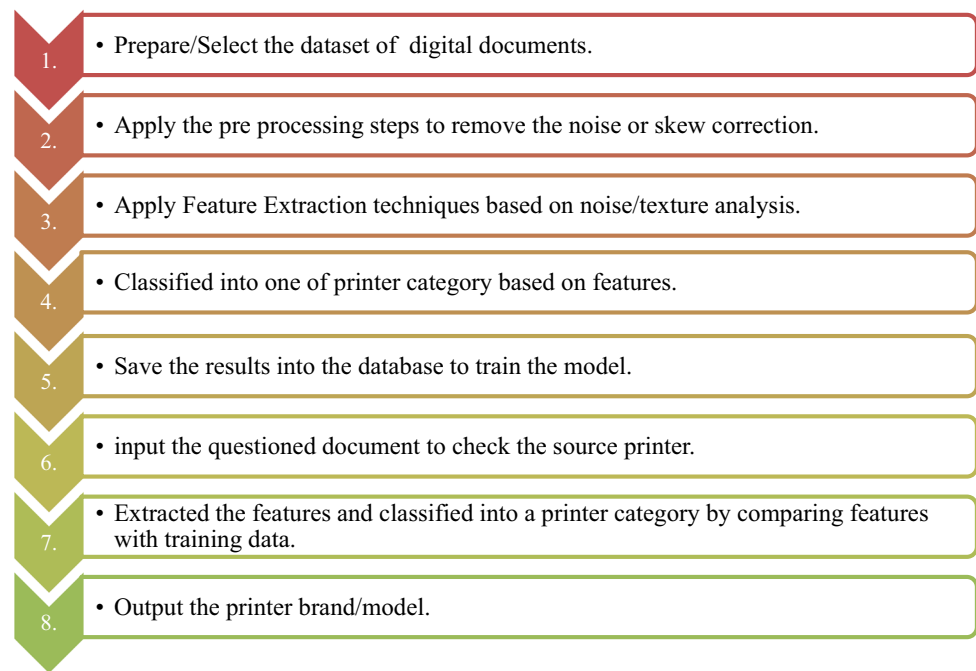
suicide notes, threatening letters, and ransom requests by analyzing handwriting and the printing process of the documents.

- **Fake insurance claims:** Document forensics aids in detecting fraudulent insurance claims by investigating medical claim bills, fake accident reports, duplicate photographs, or forged signatures from higher authorities in the medical field.
- **Property dispute issues:** In cases of trademark infringement or intellectual property disputes, document analysis is employed to examine product documentation, labeling, and packaging to identify unauthorized usage or imitation.
- **Counterfeit detection:** Document forensics helps identify counterfeit money, passports, or official documents. Investigators detect these forgeries by analyzing the printing processes, security features, and paper quality of the documents.
- **Identification of authorship:** Document forensics utilizes features of handwritten letters, threats, and ransom notes to identify their authors through handwriting analysis, linguistic research, and other methods. This is particularly useful in cases of harassment and criminal investigations.
- **Historical document authentication:** Forensic analysis aids in authenticating ancient manuscripts, letters, or objects through material examination, writing style analysis, and contextual hint analysis.
- **Divorce or custody disputes:** Document forensics helps verify the authenticity of documents such as bank records, wills, and agreements through handwriting analysis and document examination in court proceedings related to divorce or custody disputes.

2 Research methodology in document forensics

Most researchers utilize supervised machine learning techniques, which involve both a training phase and a testing phase. In the training phase, the model is trained by extracting different types of features from a given dataset and saving these features for printer classification. In the testing phase,

Fig. 4 Different steps of training and testing phase



this model is applied to unknown documents using the same feature extraction techniques, and the data are compared with the training data to determine the source printer of the document. The steps of both phases are explained in Fig. 4:

Machine learning techniques are used mostly to develop a model for document forensics. Different kinds of classification techniques are used under the supervised learning techniques like decision trees, Naive Bayes, K-nearest neighbors (KNNs), and support vector machines (SVMs). Convolutional neural network (CNN) is also used under deep learning techniques for the authentication of the document. CNN helps to identify the spatial and structural patterns of the text or the images which helps to differentiate between different sources or detect alterations. Ferreira et al. [18] utilize convolutional neural networks (CNNs) applied to various data representations, enabling the detection of nearly invisible and challenging printer footprints created during the printing process, without requiring prior knowledge of specific printing artifacts. Recurrent neural networks (RNNs) are very useful to identify the handwritten patterns of the author. RNN also helps in the analysis of ink strokes. Autoencoder is also a deep learning method of neural networks that is quite good at detecting document forgeries since they are utilized for unsupervised anomaly detection. The usage of Transformer (deep learning architecture) is increasing day by day in handwritten analysis because it can handle large and complex handwriting datasets very effectively. Moreover, combining different deep learning techniques like CNN & RNN can increase their effects and produce more accurate results.

There are certain steps that researchers must follow to implement document forensics for source printer identification. The common research methodology used in document forensics techniques is as follows:

- i. *Input questioned document*: First of all, the input must be the questioned document. The model must check the authenticity of the document whether is it real or a forged document. The document can be an invoice, literature, contract document, or any other type. Texture-based techniques can work only on text documents but noise-based techniques can work on the document including images, signatures, and other information.
- ii. *Preprocessing*: The preprocessing steps will be applied to the document based on the different techniques. Some common preprocessing steps can be color correction, noise removal, gray scaling, skew correction, compression, or decompression of the document image.
 - In the color correction, the white balance of the image and contrast can be adjusted. The color correction step is also helpful in detecting printer patterns or ink types. Gray scaling is also a part of color correction which helps to reduce the computational load and helps to apply the different feature extraction techniques like SIFT, SURF, ORB, etc.
 - The noise removal can be achieved with the techniques like Gaussian or median filtering which improves the clarity of the document. Spatial filter, Wiener filter, and Gabor filter are also used for the purpose of noise reduction.

- Skew correction helps to detect and correct the rotation in the document. Segmentation is also can be done in the preprocessing in which the document is separated into different regions such as texture part, Images part or signatures.
- Segmentation helps to focus on the necessary parts of the document removes unnecessary objects and reduces noise.

These preprocessing steps prepare document images for accurate feature extraction, classification, and other forensic analyses, allowing forensic investigators to make reliable conclusions based on consistent, high-quality data.

- Feature extraction:* After applying the preprocessing steps successfully, the next step is extracting the different kinds of features from the document which are unique of its kinds. Feature extraction techniques can be based on texting styles, noises generated by the different printers, inks used by the different printers, sensor pattern noise, image quality features, and also some handcrafted features based on speeded up robust features (SURFs), key printer noise features (KPNFs), and oriented FAST and rotated BRIEF (ORB).
- Classification:* The next step is to classify the document into one of its printer classes based on the different features extracted. Different kinds of classification techniques are used in document forensics like support vector machine (SVM), K-nearest neighbor (KNN), decision trees, random forest, Naive Bayes, etc.

3 Design constraints for this study

The availability of affordable and low-cost gadgets has led to a significant increase in the acquisition of digital photos over the past two decades. Nowadays, digital documents are used as proof to establish the legitimacy of material in various fields. However, people sometimes intentionally modify these documents to deceive viewers. The field of recognizing faked digital documents is known as document forensics. Many researchers have shown interest in this topic and have produced numerous research and review articles. The number of published papers has grown significantly over the past fifteen years, with most of them published in the last seven years. Review articles and research papers were extensively searched by title using keywords such as "document forensics" and "document forgery." Inclusion and exclusion criteria for studying document forgery techniques have been established to focus solely on relevant research by excluding irrelevant studies. The novelty and unique contributions of this study are outlined as follows:

- *In-depth analysis of techniques:* This study provides a detailed examination of the latest document forensics methods, particularly those incorporating deep learning.
- *Comparative evaluation:* It includes a comparative analysis of various techniques, focusing on datasets and accuracy levels.
- *Feature insights:* The research delves into the distinct features of documents that play a crucial role in detecting forgery.
- *Strengths and limitations:* A thorough exploration of the strengths and weaknesses of different methods is presented.
- *Practical implementation challenges:* The study highlights real-world challenges in applying document forgery detection techniques effectively.

3.1 The inclusion criteria are:

- Studies published between 2003 and May 2024 will be reviewed.
- Only studies that directly address the topic of document forensics analysis will be included.
- Studies with significant research on source printer identification will be considered.

3.2 The exclusion criteria are:

- Studies published in languages other than English will be excluded.
- Studies that are incomplete or ambiguous regarding document forgery will be excluded.
- Studies published before 2003 will be excluded.

3.3 Research questions

This study aims to identify and evaluate various document forgery detection and printer identification processes. To achieve this, several research topics are addressed. Below is a description of the study's research questions:

- What different approaches are there for detecting forged digital documents?
- Which techniques are used to identify forged documents?
- What feature extraction techniques are used to detect forgery in documents?
- What are the various challenges in implementing the detection process?
- What are the future directions for research in this area?

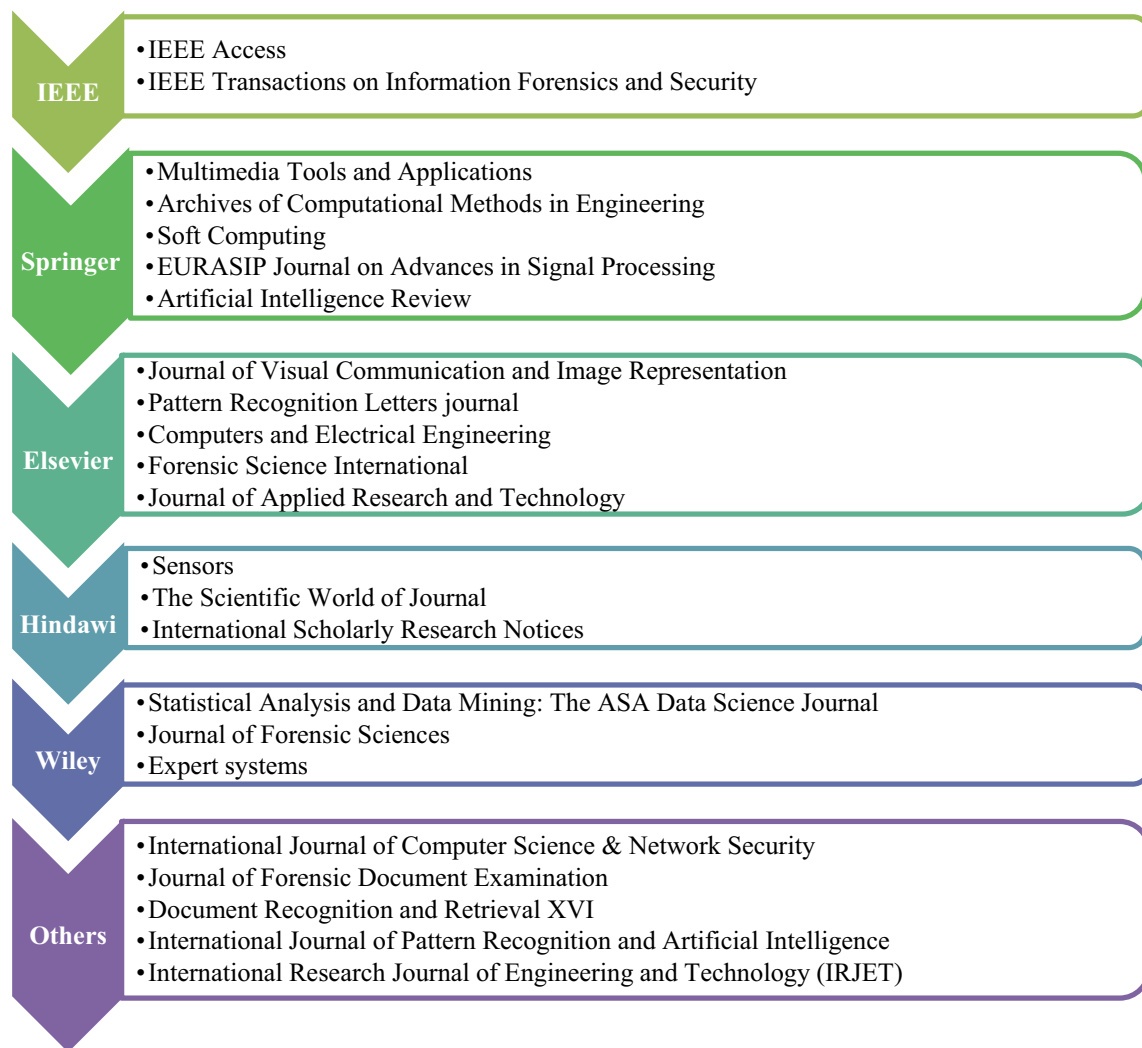


Fig. 5 List of journal

3.4 Data sources for research work

Identifying original and genuine sources for research is crucial. This review paper gathers information from various sources to provide a complete understanding of the topic. The data sources and research papers included are specifically centered on the subject of document forensics analysis. Only authentic and trustworthy sources are considered for this study. Articles and research papers have been sourced from reputable platforms such as ResearchGate, Springer, IEEE Xplore, Wiley Journal, and others. Several important journals with diverse articles are considered for analysis, as shown in Fig. 5.

easy due to the availability of inexpensive editing software. Fraudsters misuse technology to commit fraud with documents. Conversely, detecting document fraud is a challenging task. This research paper aims to raise awareness among the public and upcoming researchers about document forgery. The goal is to provide a comprehensive overview of the techniques developed or in development to detect forged documents. Many researchers have contributed to preventing forgery, emphasizing the importance of fraud detection to preserve the security and integrity of crucial records. Readers can learn more about the significance of document forgery detection and the latest techniques being developed by reading this paper.

3.5 Encouragement for readers

Document forgery is a serious problem, and cases of forgery are increasing daily. Tampering with documents is relatively

4 Reported work on document forensics analysis

This work is based on research focused on identifying the source of the document, whether it is the printer for digital documents or the actual author for handwritten documents. Feature extraction techniques primarily involve texture analysis, noise analysis, and other document features. The reported work is as follows:

Initially, Ali et al. [3] presented two strategies for source printer identification: one passive and one active. The passive technique involves distinguishing the printer by identifying unique features in the printed document. The researchers set up a specialized library containing nine printer models from six brands, tested with two different patterns, and analyzed print quality defects caused by electromechanical fluctuations, which are unique to each printer. These defects were processed using a classifier to create a decision tree for printer identification. In the active technique, an extrinsic signature is pre-embedded into the document. Experiments with four different EP printer models showed that banding effects are consistent within the same printer type but vary between different types, aiding in the differentiation of printer models. Ali et al. [4] introduced a new technique for identifying the source printer based on principal component analysis (PCA). PCA is used for dimensionality reduction, which helps create a highly effective pattern recognition tool. A Gaussian mixture model (GMM) classifier identifies the mixing coefficient for an unknown printer's projection. The technique was tested on five different EP printer models, with a sixth model used for validation, and it yielded very good and efficient results. Mikkilineni et al. [42] found that detecting the printer using banding frequency features is effective for regions with large mid tones but challenging for text. Thus, a technique was developed for small regions, such as individual text characters, for example, the character "e" was used in this research. A total of 22 features were obtained, with 20 from GLCM and 2 from variance and entropy of pixel values in the region of interest (ROI). This technique provided accurate results in detecting the printer. However, this technique requires prior knowledge of the printers; if used with an unknown printer, not in the dataset, it may be misclassified as one of the recognized printers. Khanna et al. [34] described techniques for sensor pattern noise-based image authentication of images obtained from flatbed desktop scanners. The first group of features includes statistical characteristics such as median, kurtosis, standard deviation, mean, and skewness. The correlation between each row of sensor noise and the average row is analyzed to identify similarities between the noise rows. The second set of characteristics represents the fixed portion of the pattern noise. A poor-quality scanner creates more random noise (due to factors such as lighting variations), resulting in fewer interrow correlations compared to

a high-quality scanner. This study shows that the statistical feature vector technique is more effective for source camera identification than the correlation-based strategy. Accurate results require the selection of appropriate features. An average categorization accuracy of 96% was achieved for four scanners.

Khanna et al. [35] introduced methods for identifying images taken with digital cameras, computer-generated images, and images produced by scanning. The technique relies on analyzing residual pattern noise features found in images from digital cameras and scanners. In their experiment, various classes of digital images were used, and different types of scanners with CCD or CIS sensors were tested. A total of 108 images were captured from 11 scanners at 200 dpi, 300 images were taken with three types of cameras, and numerous images were downloaded from the internet for computer-generated images. Each of the three source groups comprised 1000 images. The classification accuracy for computer-generated images compared to camera images was 91.5%, while the accuracy for camera images compared to scanner images was 97.6%, and for scanner images compared to computer-generated images, it was 89.4%. The overall average categorization accuracy across all three groups was 85.9%. By leveraging variations in image sensor geometry and the statistical properties of residual noise, the approach demonstrated promising results. Schulze et al. [53] examined edge-based gray-level features for identifying high-throughput printing techniques. Printouts from 13 distinct inkjet printers and 49 different laser printers were evaluated. These prints were created using a template from a document with a layout typically found in offices. Factors such as paper quality, ink type, and document aging were not considered. The examination took into account the scan resolution limitations that often affect high-throughput scanning systems used for document management systems (DMSs). The evaluation revealed that it is possible to recognize printing techniques even from low-resolution scans, which are unique to these systems. Ryu et al. [50] proposed a technique for recognizing halftone textures in high-resolution scanned colored documents. A classifier for document forensics was developed based on image quality measurements, utilizing a total of 17 image quality measures. The results from each measurement served as training and testing parameters for the SVM classifier. The technique was tested on a dataset of 9000 images, all printed by nine electros photographic (EP) printers, achieving an accuracy of 80%. Choi et al. [11] introduced a novel method for identifying color laser printers for halftone images using statistical analysis. The RGB domain of the printed color images was scanned and then converted to the CMYK domain. Wavelet analysis was employed to characterize the color noise in the transformed images. Initially, 39 noise features were extracted from the color-printed pictures. These features were used to train a support vector

machine (SVM) to classify the color laser printer. The experiment involved nine color laser printers from four different brands: Konica, Canon, Xerox, and HP. The approach was tested for categorizing the color laser printer brand, model, and color toner. The accuracy was 97.89% for brand identification, 92.28% for color toner, and 80.24% for printer model identification.

Schreyer et al. [52] presented a new frequency-domain method for recognizing document printing techniques, which utilizes the quantity and distribution of frequencies present in a document to create a distinct fingerprint for each printing technique. This method employs machine learning techniques and discrete cosine transform (DCT) coefficients. It can differentiate between documents printed by inkjet or laser printers and detect early photocopies at low scan resolutions. The method retrieves statistical features from the wavelet image at multiple resolutions, the DCT image, the gradient image, and the noise image. Using SVM classification and DCT frequency analysis, the method achieved the highest classification accuracy, with results of 92.92% at 400 dpi and 99.08% at 800 dpi. Van et al. [65] proposed a method for detecting combined skew and orientation in Roman script text lines using geometric modeling. This approach explores a skew range encompassing all four possible orientations for text lines. The model with the best fit provides the estimation of page skew and orientation. The dataset for the experiment included 159 images. This approach outperformed recent methods in tests using both the UW-I dataset and a custom dataset with documents rendered at various resolutions. The proposed method achieved an accuracy of over 99% on this dataset and demonstrated reliability in detecting page orientations for documents generated at 150, 200, 300, and 400 dpi. Additionally, the method is robust against changes in scanning resolution and offers the advantage of providing both skew estimates and orientation in a single process. Jiang et al. [27] proposed a method for determining the brand and model of a printer using printed and scanned images. The researchers employed a forensic technique based on Benford's law to analyze the distribution of the first digits in multi-size block DCT coefficients, which were then input into a support vector machine (SVM) classifier. This technique was found to be a reliable forensic feature because it is independent of the image's content. Additionally, the feature vector proved to be robust, meaning that random noise had minimal impact on the forensic characterization and combinational signature. By utilizing Benford's law features for printer identification, the method can represent each printer with as few as nine numbers. Experiments with electrophotographic (EP) and desk jet printers yielded average classification rates of 96.0% for five different printer brands and 94.0% for six different printer types within those brands. Mikkilineni et al. [41] developed a method for identifying the source printer by extracting the intrinsic features specific

to each printer. An intrinsic signature is difficult to erase or counterfeit because it is closely linked to the electromechanical characteristics of the printer.

Tsai et al. [62] subsequently applied the techniques to Chinese characters. The combination of GLCM statistics and wavelet transform features also yielded strong results for Chinese characters. Using an SVM classifier, they achieved a classification accuracy of over 98% on a dataset of 3750 Chinese characters. The research conducted by Bertrand et al. [7] focuses on detecting fraud in binarized, poor-resolution documents. The technique is based on comparing character shapes and identifying structural issues within the document. The study emphasizes character-level intrinsic features computed from the source material, analyzing them in two ways. The first method compares character shapes to identify similar characters, while the second method measures the distance between character features to detect flaws introduced by the forgery process. The software was developed to create fake document images, and the forgery detection technique achieved recall and accuracy values of 77% and 82%, respectively. Gebhardt et al. [20] used differences in edge roughness to distinguish between laser-printed and inkjet-printed pages. They described a novel method for document authentication that combines unsupervised anomaly detection with features of the printing technique. A key advantage of this method is that it requires no prior training with genuine documents. Additionally, they created a dataset comprising 1200 document images from various domains (contracts, scientific articles, bills), produced by thirteen laser printers and seven inkjet printers. Shang et al. [55] proposed a method to differentiate documents printed by three common types of document-producing devices: electrostatic copiers, laser printers, and inkjet printers. This technique relies on detecting features extracted from the characters. For testing algorithm performance, ten laser printers, six inkjet printers, and nine electrostatic copiers were used, with ten pages of English words and letters printed from each printer. This method achieved an accuracy of 90% and is also effective with JPEG compression.

Tsai et al. [63] implemented a general approach to extract key features for identifying the source of a printer. The technique employs various filters, including the Wiener filter, Gabor filter, and spatial filter, in conjunction with gray-level co-occurrence matrix (GLCM) and Discrete Wavelet Transform (DWT). This method was developed specifically for Japanese characters. Compared to other GLCM methods, which have an identification rate of 77.31%, the average accuracy achieved with this technique was significantly higher at 94.23%. Ferreira et al. [19] proposed several solutions for determining the source printer of a document, introducing three distinct methods that investigate intrinsic signatures. The first solution involves two descriptors based

on the multidirectional and multi-scale characteristics of texture micro-patterns found in text letters or regions of interest, focusing primarily on the inner parts of the letters. The second descriptor, known as the convolution texture gradient filter (CTGF), is based on histograms of low-level gradient-filtered patterns. This method filters the interior and exterior portions of printed letters and figures using filters of one or more scales. The third solution allows for determining the document's source even if some portions are incorrect or missing. When the complete document is available, combining this approach with integrated strategies can yield even more reliable results. The accuracy rates for characters, frames, and documents with the first method were 97.60%, 98.38%, and 88.58%, respectively. The results indicated that the accuracy for documents was 88.45%, and for frames, it was 94.19%. Following this experiment, a new dataset was created and made freely available for further research.

Ferreira et al. [18] proposed a novel deep learning-based method for attributing laser printers, focusing on systems that can identify patterns in printing directly from scanned copies of printed documents without additional feature engineering. This method utilizes convolutional neural networks (CNNs) applied to various data representations, enabling the detection of nearly invisible and challenging printer footprints created during the printing process, without requiring prior knowledge of specific printing artifacts. The method achieved an overall accuracy of 97.33% across all experiments. Joshi and Khanna [30] introduced a new approach for categorizing the source printer from scanned images of printed documents by utilizing all printed letters simultaneously. The proposed method was tested on two new datasets: one consisting of 18 printers scanned at 600 dpi and 300 dpi, printed in four different fonts, and another with ten publicly available printers. This approach reduces the number of printed pages needed for training. In the dataset with 18 printers—three of which are identical in make and model—the method achieved an accuracy of over 99% with just 5 pages per printer. Tsai et al. [61] proposed a machine learning-based technique for analyzing microscopic images of both text and picture documents. This method employs SVM classification to determine the source of laser printers by extracting features such as gray-level co-occurrence matrix (GLCM), local binary pattern (LBP), discrete wavelet transform (DWT), Gabor filter, Wiener filter, spatial filters, Haralick features, and SFTA. When using LBP features, the identification success rate can reach 99.89%, demonstrating the technique's high classification accuracy and its potential for real-world digital forensics applications. Kim et al. [37] also introduced a technique to identify color laser printers based on cascaded learning of deep neural networks.

Khan et al. [33] introduced an automatic technique utilizing fuzzy C-means clustering to detect ink mismatches. This technique clusters the spectral responses of ink pixels

in handwritten notes into distinct groups, corresponding to unique inks used in the document. The method was tested using the publicly available UWA Writing Inks Dataset and showed improved results compared to other methods. Tsai and Yuadi [61] have also worked on font textures, including not only English characters but also Chinese, Japanese, and Arabic characters. Dorem et al. [13] proposed a technique that operates directly on compressed digital images to detect forgery. This supervised technique employs two algorithms, copy-move and copy-paste, and was tested on two datasets: 260 forged images and 300 JPEG-compressed official mark cards. Gupta and Kumar [21] also used a passive approach to identify the source of the printer by leveraging handcrafted features based on speeded up robust features (SURFs), key printer noise features (KPNFs), and oriented FAST and rotated BRIEF (ORB). They utilized classifiers such as KNN, decision trees, random forests, and majority voting. To manage the large memory requirements of SURF and ORB descriptors, K-means clustering and locality preserving projection (LPP) were employed. By integrating adaptive boosting and bootstrap aggregating techniques, the proposed system achieved a 1.7% increase in accuracy, reaching 95.1% with the random forest classifier and adaptive boosting approach. Kumar et al. [39] developed a classifier-based method for determining the source printer and categorizing documents by printer type. Their novel approach uses key point-based feature extraction with ORB, SURF, and BRIEF descriptors. Combining Naive Bayes, KNN, and random forest classifiers with a majority voting mechanism and adaptive boosting techniques, they achieved an accuracy of 86.5%. Sirajudeen et al. [57] proposed a technique utilizing convolutional neural networks (CNNs) in conjunction with optical character recognition (OCR) and local binary patterns (LBPs) for detecting document forgery. The approach also employs oriented FAST and rotated BRIEF (ORB) to extract images from documents. The model was trained using the MIDV-500 dataset, which includes 256 Azerbaijani passport images. This method effectively analyzes both text and images, such as seals, stamps, or holograms. When compared to existing models, the CNN-based forgery detection method shows promise for real-time application implementation.

Hamzehyan et al. [22] developed a technique based on texture analysis to identify the source printer. Their novel method utilizes LBP features to extract primary features, which are then used to derive secondary features through joint factor analysis. The digital document is divided into patches, with LBP features extracted from each section. This technique achieved an accuracy of 98.48%. By relying solely on a single low-dimensional i-vector feature for each document image and avoiding the use of OCR algorithms, this method reduces the computational burden of the classification process and eliminates the need for majority voting. Moreover,

the technique is not limited to any specific language or character set. Nandanwar et al. [45] developed a technique to identify altered text in documents. This method relies on DCT coefficients and their positive and negative relationships, which help detect distortions in the document. Laplacian filtering is used to identify the growing differences between text values and other pixels. The results are highly effective and represent a significant improvement over existing techniques for detecting altered text. Djeddi et al. [14] proposed a method to support the theory that the codebook primarily serves as a representation space for comparing various writings, with the patterns in the codebook having minimal impact on identification and verification performance. They created several codebooks using handwritten examples in Greek, Arabic, and Chinese to test this hypothesis.

Joshi et al. [29] analyzed and proposed a technique based on printer-specific pooling, which improves the performance of printer-specific local texture descriptors across two datasets. Instead of using a complex machine learning-based classifier, their method adopts a straightforward correlation-based prediction approach. They also introduced a printer-specific local texture descriptor (PSLTD) for identifying the originating printer from scanned documents. When trained only with documents in the Cambria typeface, the method achieved average classification accuracies of 93.5%, 94.3%, and 60.3% for Arial, Times New Roman, and Comic Sans fonts, respectively. Ho et al. [47] suggested a source printer identification approach that extracts features from printed microscopic patterns using convolutional neural networks (CNNs). Various classification methods, including multi-class support vector machines and random forest, are employed to process these features. To evaluate effectiveness, actual printed dot patterns from widely used printing techniques, including electrophotography, waterless offset, and conventional offset, were used. This study highlights the importance of grayscale patterns and CNN-based features in determining the printer's origin. The hybrid classification algorithm achieved an accuracy of approximately 99% on the test set, surpassing previous research that relied on manually developed features. El Abady et al. [17] proposed a technique based on global features such as the histogram of oriented gradient (HOG) and local features such as local binary pattern (LBP) descriptors to find out the source printer class. This experiment was done on a dataset of 1200 documents printed from 20 distinct printers. For the classification techniques, decision trees (DTs), k-nearest neighbors (KNNs), random forests, aggregate bootstrapping (bagging), adaptive boosting (boosting), support vector machine (SVM), and mixtures of these classifiers have been used. This method provides an accuracy of 96% with adaptive boosting.

Bibi et al. [8] reviewed research on source printer identification, comparing text-dependent versus text-independent studies. They conducted an experiment using a standard

dataset with documents printed from 20 different printers. The accuracy achieved was 95.52% for text-independent patches and 98.06% for text-dependent characters. Previous techniques primarily focused on pre-segmented characters and calculated identification rates by matching identical characters. This technique, however, applies to entire printed document images and reports a higher identification rate. Ghadi et al. [2] proposed a model based on guilloche patterns in forged documents to detect forgery. The model operates in two steps. The first step is feature extraction, which uses a convolutional neural network (CNN) architecture to learn and extract discriminative features from two documents. The second step involves measuring similarity to determine whether the document is authentic or forged based on these features. This technique was implemented using the MIDV-2020 dataset. Recent studies have identified machine learning, deep learning, autoencoder-decoder architectures, and convolutional neural networks as some of the most popular techniques for obtaining, analyzing, and utilizing data derived from hyperspectral document images. Hyperspectral images are produced by capturing images at hundreds of wavelengths to collect more data, including depth information [69]. Jaiswal et al. [25, 26] presented an unsupervised deep learning method utilizing spectral features for ink matching in hyperspectral document images. The method employs a convolutional autoencoder (CAE) to extract features and applies logistic regression (LR) to identify ink mismatches. The publicly available UWA Writing Ink Hyperspectral Images (WIHSI) dataset was used for experimentation, featuring seven individuals who used five different black and five different blue inks. The CAE-LR approach achieved a classification accuracy of 99.05% for blue inks and 92.74% for black inks. Wang et al. [67, 66] suggest a self-supervised blind-block network (called BockNet) with a guard window in order to create an excellent background reconstruction network for HAD (hyperspectral anomaly detection) applications. When reconstructing the central pixel, BockNet places a blind-block (guard window) in the middle of the network's receptive field, making it impossible for the network to perceive the data inside the guard window. Cheng et al. [10] propose a unique deep feature aggregation network (DFAN), which creates a new paradigm for HAD to represent numerous background patterns. This technique enables the efficient use of spatial and spectral data to record the background and anomaly distribution. A hierarchical integration framework for hyperspectral simultaneous anomaly detection (AD) and denoising (HyADD) is proposed by Wang et al. [68] to address the issue of hyperspectral images being corrupted by additive noises, which makes it difficult to precisely observe and identify the materials on Earth's surface. A novel frequency-to-spectrum mapping generative adversarial network for HAD is proposed by Wang et al. [67, 66]. In order to map and restore the

backgrounds from the FrFD to the original spectral domain, FTSGAN must learn the embedded characteristics of the backgrounds due to the semisupervised technique.

Jaiswal et al. [25, 26] introduced another hyperspectral image-based technique. This method, based on CAE and LR, incorporates spectral bands, spatial patches, and spectral-spatial patches. The dataset used was the same as in the earlier technique. The results demonstrated that including spectral and spatial patches improves classification accuracy: for black inks, accuracy increased by 4.85%, and for blue inks, by 0.13%. Im et al. [24] presented one of the first techniques to identify the printing technology of historical images. This work is distinct from computer vision models because it requires the use of microlevel characteristics. Lee et al. [40] presented a new approach for paper identification based on hybrid features. This technique combines texture features extracted from digital documents using two methods: gray-level co-occurrence matrix (GLCM) and convolutional neural networks (CNNs). The study used seven types of paper brands and achieved an accuracy of 97.66%. This method confirms the technique's utility for paper products and its effectiveness in solving criminal cases of document forgery.

A lot of work has been done recently on handwritten documents. Handwritten document analysis is a part of forensic science that aims to identify the actual writer of a handwritten document based on intrinsic features such as text line heights, spacing between words, and character sizes using image processing and deep learning techniques [9]. Crawford et al. [12] introduced a model with a statistical approach to identify the author of a handwritten document, which also works for cursive writing styles. Najla et al. [44] developed a novel technique based on graphology to detect gender from handwritten documents. SVM and ANN classifiers were used in this technique, achieving accuracies of 94.7% and 97.1% with SVM and ANN, respectively. Hazra et al. [23] developed a semiautomated tool named DIGIDOC to analyze handwritten documents and verify signatures. Developed in Python, the tool features a user-friendly interface and includes automatic preprocessing, noise removal, and signature verification. Tyagi et al. [64] researched handwritten documents using hyperspectral imaging to prevent document forgery. The study investigated hyperspectral imaging (HSI), which provides complete spectral information about a scene, helping to detect small spectral changes in handwritten documents. The research employed the CAE-SVM technique, where a convolutional autoencoder extract features and SVM serves as the classifier. Tested on the publicly available UWA Writing Ink Hyperspectral Images dataset with blue and black inks, this method achieved an accuracy of 92.88% for blue ink, outperforming other existing methods and technologies. The HSI imaging approach has applications in agriculture, environmental monitoring, remote sensing, forensics, document analysis, and medical imaging.

El-Kenawy et al. [15, 16] introduces the graylag goose optimization (GGO) algorithm, a swarm-based nature-inspired optimizer inspired by geese's efficient group flight patterns. Tested on UCI datasets and engineering benchmarks. The Puma Optimizer (PO) is a meta-heuristic algorithm inspired by puma behavior, designed for efficient problem-solving across diverse optimization challenges. It outperforms other algorithms on multiple benchmarks and machine learning tasks, proving particularly effective in clustering, feature selection, and community detection [1]. El-Kenawy et al. [15, 16] presented a new meta-heuristic inspired by football strategies to tackle complex optimization problems. By mimicking tactical plays, FbOA balances exploration and exploitation, showing superior convergence, accuracy, and robustness in benchmark tests compared to other leading algorithms.

Mirza et al. [43] proposed a unique method for extracting signatures from documents using hyperspectral imaging. Extracting accurate signatures can be challenging due to overlaps with stamps, typed text, or other document parts. The method uses the pure pixel index approach for specific overlapping regions and spectral classification methods to separate the overlapping signatures. The results are effective in recovering signatures from documents with high accuracy. Kim et al. [38] introduced a new technique for handwritten documents using deep learning algorithms. The method involves bootstrapping to divide the data into small parts and applying a transfer learning algorithm to extract features as latent vectors. Cosine similarity between these latent vectors measures the similarity between two handwritten documents. The findings indicate that the technique can accurately categorize document pairings as belonging to different authors or the same author in most cases (Table 1).

4.1 Analysis based on the reported work

The objective of this study is to give a comprehensive overview of the methodology, dataset, and performance measures of the techniques used in source printer identification. Table 2 presents a summary of the different techniques along with the datasets and their accuracy levels.

5 Critical analysis based on survey

A detailed discussion of the various authors' approaches was covered in the last section. Section 5 has previously provided the whole step-by-step procedure for the forgery detection process. Following a thorough analysis of published work, it was shown that the three primary phases determine a method's overall effectiveness. These phases are dataset selection, feature extraction techniques, and classification techniques.

Table 1 Comparing active and passive forgery detection techniques with its different parameters

Aspect	Active techniques	Passive techniques
Definition	Involve embedding information (e.g., watermarks or signatures) into the document to detect document fraud	Analyze inherent features or inconsistencies in the document without prior embedded information
Strengths	High accuracy when embedded information is intact	No need for prior modification of the document Effective for analyzing legacy or unmarked documents
Weaknesses	Easy detection of tampering if watermarks/signatures are robust Requires modification of the original document Can be circumvented if embedded data are removed or altered Not suitable for legacy or unmarked documents	Utilizes natural characteristics of the document, such as texture, noise, or format inconsistencies Relatively lower accuracy for complex forgeries Heavily reliant on sophisticated algorithms Prone to false positives if intrinsic features are misunderstood
Applications	High-security environments where embedding is feasible (e.g., financial, government)	Forensics, legal investigations, and applications involving unmarked documents
Example techniques	Digital watermarks QR codes or cryptographic signatures	Printer identification using texture analysis Deep learning models for forgery detection
recent trends	Enhanced robustness of watermarks through cryptographic techniques	Increasing use of machine learning and deep learning to identify complex patterns
Cost of implementation	Higher due to embedding process and potential need for specialized hardware/software	Generally lower but requires computational resources for advanced algorithmic analysis

Feature extraction techniques play a significant role in document forensics. In recent years, deep learning models have been developed for feature extraction. Different kinds of feature extraction techniques (Fig. 6) are based on:

- Texture patterns analysis of the document.
- Unique noise patterns are generated by the printer.
- Boundaries/edges and outline analysis of the document.
- Ink mismatch analysis of the document printed by different printers.
- Frequency-domain analysis by transforming the document.
- Handwritten document or signature analysis.
- Features extracted by OCR (optical character recognition) from the document.
- Spatial and structural analysis of the document.

As shown in Fig. 6, there are certain types of features of the documents or the printers which help to detect the forgery in the document.

These features contain unique information which different kinds of classification techniques are used in document forensics such as SVM, decision trees, random forest, k-nearest neighbor, Naïve Bayes, ANN, CNN and K-means clustering. SVM is the most common technique used by the researchers. SVM is a supervised machine learning algorithm that is used for multi-class classification, particularly in high-dimensional spaces. In document forensics, SVM is a

dependable, accurate, and interpretable option due to its generalization capabilities, efficacy with high-dimensional data, and capacity to handle small samples. By using these classification techniques, documents can be classified accurately for source printer identification, fraud detection, and authentication of the document. The robustness of document forensic analysis is further increased by applying a combination of these classification techniques.

Specific features play crucial roles in forgery detection by capturing distinct document characteristics. GLCM analyzes texture by examining pixel intensity relationships and identifying inconsistencies in printer patterns. LBP encodes micro-texture variations, aiding in detecting tampering in document surfaces. Discrete cosine transform (DCT) identifies frequency-domain anomalies, useful in uncovering compression artifacts from digital forgeries. SIFT (scale-invariant feature transform) extracts robust key points, enabling the detection of localized manipulations like altered signatures. SURFs (speeded up robust features) provide faster key point extraction, ideal for large-scale forgery detection. Combined, these features enable powerful analysis, especially when integrated with machine learning models.

5.1 Comparative analysis of different techniques:

As we have already discussed the different kinds of feature selection techniques, different kinds of classification techniques and also the different kinds of features used in

Table 2 Comparative analysis of different techniques

Author and year	Features based on	Feature extraction technique	Dataset	Classifier	Techniques used	Results	Remarks
Ali et al. [3]	Texture-based	Banding frequencies	4 Different models of EP printers	Decision Tree classifier	Passive approach Active Approach	Good and efficient results	This technique is not suitable for text-only documents
Ali et al. [4]	Other Based	Principal component analysis (PCA)	5 EP Printer for training & 1 EP for testing	GMM (Gaussian mixture model) Tree Classifier	CART (Classification & regression Tree)	Good Results	This technique is suitable for unknown printers
Mikkilineni et al. [42]	Texture-Based	Graylevel co-occurrence texture (GLCM)	10 EP Printers of different Brands and Models	5NN classifier	1. Texture feature-based descriptors	Total 22 features obtained	This technique is not suitable for unknown printers
Khanna et al. [34]	Noise Based	sensor pattern noise-based image authentication	Total 1200 images from 4 scanners	SVM Classifier with feature vector based	median, kurtosis, standard deviation, mean and skewness	96% Accuracy for Average classification	This technique is not suitable for lower resolution images
Khanna et al. [35]	Noise Based	sensor pattern noise	Total 1000 images	SVM Classifier	Sensor pattern noise	The accuracy is 85.9%	Total 15 features extracted from image
Ryu et al. [50]	Text Based	Image Quality Features	9000 Images	SVM Classifier	17 image quality measures developed	Accuracy of 80%	–
Schulze et al. [53]	Texture-based	Edge based Gray-Level-Features	Documents from 13 inkjet and 49 laser printers	Decision Tree, SVM, Multilayer Perceptron	Perimeter Based Edge Roughness Distance Map Edge Roughness Gray Value Distribution	1. 75–80% for decision tree 2. 60–70% for SVM	The constraints of high-throughput systems are also capable to recognize the printing technique of a document
Choi et al. [11]	Noise Based	Color Noise Features	99 digital images	SVM Classifier	Wavelet analysis	Accuracy: 97.89% -brand, 92.28% toner, 80.24% model of color laser printer	39 noise features are extracted from color-printed pictures

Table 2 (continued)

Author and year	Features based on	Feature extraction technique	Dataset	Classifier	Techniques used	Results	Remarks
Schreyer et al. [52]	Other Based	DCT (Discrete Cosine Transformation) Features	49 laser-printed, 14 inkjet-printed, 46 photocopied documents	SVM Classifier	1. DCT 2. Printing Process Characteristics	The accuracy at 400 dpi- 92.92% 800 dpi-99.08%,	This approach is also suitable for the high-throughput document processing
Van et al. [65]	Other Based	Page Skew and Orientation Detection	979 binarized images of medical journal articles	–	Geometric Text-Line Model One-Step Skew Orientation Detection	99% accuracy	Best part of this technique is to obtain the skew and orientation both in a single step
Jiang et al. [27]	Noise Based	Multi-size Block Benford's Law	150 high-resolution images	SVM Classifier	1. Printer Principle and Half toning 2. Multi-size Block DCT Coefficients Statistics	Accuracy: 96.0%-printer brands, 94.0% -printer models	This technique further can be used with the both forensics features and noise features to identify printer models
Tsai et al. [62]	Texture-based	GLCM statistics along with wavelet transform features	The dataset is used of 3750 Chinese characters	SVM Classifier	GLCM (gray-level co-occurrence matrix), wavelet transform features	Achieved the accuracy over 98.64%	This technique is best suitable to identify the Laser printers
Bertrand et al. [7]	Texture-based	Intrinsic features at character level	Three types of dataset used with 20,000 number characters	–	1. Comparing character shapes and looking for structural issues	The accuracy for recall is 77% where 82% for precision	This technique can also be used noise estimation model to give more effective results
Gebhardt et al. [20]	Other Based	Intrinsic features based on the difference in edge roughness	1200 document images	1. Grubb's Test 2. KNN Classifier	Grubb's Test KNN Tesseract-OCR	This Technique gives best outlier rank score	This is an unsupervised technique so no prior knowledge is needed about the source of documents

Table 2 (continued)

Author and year	Features based on	Feature extraction technique	Dataset	Classifier	Techniques used	Results	Remarks
Shang et al. [55]	Texture-based	features extracted from individual characters	ten laser printers, six inkjet printers, and nine electrostatic copiers used	SVM Classifier	Noise energy, contour roughness, and average gradient	Accuracy of 90%	This is the first study to differentiate between papers created using an electrostatic copier, an inkjet and a laser printer
Ferreira et al. [19]	Texture-based	multidirectional, multi-scale and low-level gradient texture patterns	120 document printer by 10 laser printers yielding 1184 TIFF images	SVM Classifier used. Feature vectors are also used to train classifier	1. Convolution texture gradient filter 2. Investigation of texture artifacts	Accuracy 97.60% for characters, 98.38% for frames and 88.58% for documents	The dataset used in this research is freely available through FigShare with source code on Github
Tsai et al. [63]	Texture-based	GLCM statistics along with Wavelet Transform Features	12 different kinds of printers with 500 “シ” letter images	SVM Classifier	1. Spatial filter 2. Wiener filter 3. Gabor filter	The average accuracy is 94.23%	–
Ferreira et al. [18]	Texture-based	Data-Driven Feature Characterization Techniques	Total 1184 images	1 Feature vector 2. Classification With Early and Late Fusion	1. Convolutional Neural Networks (CNNs)	This method gives 97.33% overall accuracy	–
Joshi and Khanna [30]	Texture-based	Local texture pattern-based features	Publicly available of 10 printers 18 printers printed in four different fonts	SVM Classifier	1. Morphological Filtering	Accuracy more than 99% achieved in the dataset 2	This method is also useful for low-resolution scanners and also the documents scanned by smartphones
Tsai and Yuadi [61]	Texture-based	Local binary pattern (LBP) features	10 sets of images from microscopic image database of 12 printer sources	SVM Classifier	GLCM, DWT, Gabor filter, Wiener filter, spatial filters, SFTA features	This method achieves the 99.89% accuracy	This research doesn't use the scanner to digitize the image but analyzes the microscopic

Table 2 (continued)

Author and year	Features based on	Feature extraction technique	Dataset	Classifier	Techniques used	Results	Remarks
Khan et al. [33]	Ink Mismatches	Fuzzy C-means clustering	UWA Writing Inks Dataset	–	Sauvola's local thresholding technique	provides better ink discrimination results than state-of-the-art methods	an efficient automatic ink mismatch detection technique
Gupta and Kumar [21]	Other Based	Handcrafted features based on KPNF, ORB & SURF	50 documents per printer printed by total 20 inkjet & laser printers	KNN, decision tree, random forest, majority voting	1. Adaptive boosting 2. Bootstrap aggregating	This method achieves the 95.1 % accuracy	Adaptive boosting & bootstrap aggregating techniques increase the accuracy by 1.7%
Kumar et al. [39]	Other Based	Key point-based features like SURF & ORB Features	50 documents per printer printed by total 20 inkjet & laser printers	1. Naive Bayes 2. KNN 3. random forest	1. Majority Voting scheme 2. Adaptive boosting	This method achieves the 86.5% accuracy	This technique gives much better results as compare to the previous work
Darem et al. [13]	Other Based	Copy Move & Copy Paste	Two datasets with 260 forged images and 300 JPEG official marks cards	–	Quantized/ Quantized Coefficient AC Coefficient & DC Co efficient	Quite accurate results as compare to the other techniques	Best technique to work on the compressed images
Sirajudeen et al. [57]	Texture-based	Optical character recognition (OCR) and local binary patterns (LBPs)	MIDV-500 dataset with 256 Azerbaijani passport images	–	oriented FAST and rotated BRIEF (ORB) (LBP)	–	CNN-based forgery detection method shows promise for real-time application implementation
Hamzehyan et al. [22]	Texture-based	Local binary pattern (LBP) features	Total 10 types of printers are used	–	joint factor analysis low-dimensional 1-vector feature	Accuracy of 98.48%	This method can be used for multiple languages or character sets
Joshi et al. [29]	Texture-based	PSLTD (printer-specific local texture descriptor)-based feature extraction	Dataset 1: 1184 Wikipedia pages Dataset 2: 720 pages	1. Correlation based prediction method	1. Location-based pooling technique	Accuracy: 93.5% for Arial, 94.3% in Times New Roman, and 60.3% in Comic Sans font types	This method is not so much suitable for intra-page forgery locations

Table 2 (continued)

Author and year	Features based on	Feature extraction technique	Dataset	Classifier	Techniques used	Results	Remarks
El Abady et al. [17]	Noise-based	Histogram of oriented gradient (HOG) and local binary pattern (LBP) descriptors	A dataset of 1200 papers from 20 distinct (13) laser and (7) inkjet printers	Decision trees, k-nearest neighbors, random forests	Aggregate bootstrapping (bagging), Adaptive boosting	Accuracy of 96%	This algorithm gives better results as compared to the recent techniques which used the same dataset
Phan-Ho et al. [47]	Other-based	Convolutional neural network	Total 4800 sample documents	SVM classifier random forest classification	Conventional Offset watermarkless offset Electro photographic	Accuracy of more than 99%	Microscopic patterns are very useful for identifying different printing sources
Ghadi et al. [2]	guilloche patterns based	1.convolutional neural network architecture	MIDV-2020 dataset	–	1.similarity measure step	Quite accurate results	The implementation code and the dataset is also publically available
Bibi et al. [8]	Texture-based & Patch-based	deep visual features	20 different printers	SVM Classification	–	Accuracy of 95.52% and 98.06% are reported using patches and characters	The suggested method claims high identification rates and operates on full pictures of printed documents
Jaiswal et al. [25, 26]	Based on Ink Mismatch	Convolutional autoencoder (CAE)	UWA Writing Ink Hyperspectral Images (WIHSI) dataset	Logistic Regression (LR)	1.logistic regression (LR) 2. spectral bands, spatial patches, and spectral-spatial patches	Accuracy of 99.05% for black ink and 92.74 for black ink	This is the first unsupervised learning technique to find the ink mismatching and gives best results compared to other techniques

Table 2 (continued)

Author and year	Features based on	Feature extraction technique	Dataset	Classifier	Techniques used	Results	Remarks
Jaiswal et al. [25, 26]	Based on Ink Mismatch	Convolutional autoencoder (CAE)	UWA Writing Ink Hyperspectral Images (WIHSI) dataset	Logistic regression (LR)	Logistic Regression (LR)	Increases accuracy by 4.85% for black inks, 0.13% for blue inks	This is the first unsupervised learning technique to find the ink mismatching and gives best results compared to other techniques
Lee et al. [40]	Texture-based	1. GLCM 2. CNN	7 types of paper brands	–	GLCM CNN	Accuracy of 97.66%	The study is very effective for the identification of paper products
Tyagi et al. [64]	Texture analysis of handwritten documents	Convolutional autoencoder	UWA writing ink hyperspectral images dataset	SVM Classifier	hyperspectral imaging (HSI)	Accuracy of 92.88%	Best suited technique for hand written documents
Mirza et al. [43]	Pixel-based identification	Hyperspectral Imaging	59 HIS Images	Spectral classification	pure pixel index approach	This approach can precisely retrieve the signature	Best suited technique to extract accurate signatures
Kim et al. [38]	Character-level features of handwritten documents	Latent feature vectors	1. CSAFE handwriting data 2. CVL dataset	–	Bootstrapping Transfer learning	Highest accuracy of 97.15%	The methodology is easily adjustable to various uses and comes with few serious limitations

Fig. 6 Various kinds of features used in document forensics



document forensics, here some points are listed related to the strengths and weaknesses of the techniques:

- CNN is one of the best deep learning techniques because of its extracting spatial features from large datasets but the main weakness is computational complexities which makes it very difficult to implement.
- Decision tree is a very effective classifier for small datasets but not much effective with big datasets and very less robust with noise.
- SVM is also a very effective technique for high-dimensional data with limited datasets but not suitable for large datasets and less interpretable as compared to techniques like decision trees.
- Naïve Bayes is quite fast and suitable for large datasets but not so much effective for images forensics because of its assumption feature independently.
- Handwriting recognizing feature is very suitable for identifying the signatures on the handwritten documents, but the results are dependent on the scanning quality which may lead to the wrong results.
- Fourier transform methods are used to analyze the frequency components of the digital documents which help in identifying printer-specific frequency artifacts but require special tools to implement the Fourier transformation and the results are also based on the resolution of the document.
- Features related to ink mismatches are highly effective in identifying ink types and the altered text in the document, but the use of this technique is very limited in document forensics.
- Combining the multiple models like boosting and bagging techniques also gives better results as compared to the single technique, but the major issue is that combining the models increases the computational stress on the model, and it is very hard to interpret the data and the results.
- Random forest is also a better technique than decision trees because of its reduced over fitting and increased accuracy. It is also very effective for the large datasets of document forensics. However, the main weakness is its interpretation is very difficult because of the combination of different trees.

This comparison demonstrates how document forensics gains from a broad range of methods, each of which is suited to particular document kinds or forensic objectives. The sensitive nature of documents and confidentiality issues in sectors such as finance and law limits the amount of publicly available datasets. Variations in document types, printing methods, and forgery techniques are not represented in many of the accessible datasets due to their lack of diversity. When the datasets become more standardized and available, hybrid methods and deep learning techniques hold the potential for increasing accuracy. Hybrid approaches combine the flexibility of passive analysis with the durability of active techniques, such as watermarks. Blockchain-integrated systems and machine learning models with watermarks are recent examples of safe, multilayered verification.

6 Future scope and conclusion

The deployment of these techniques in the real world comes with a lot of practical challenges. The major challenges can be the technical limitations, awareness about the fraud documents, document quality issues, and the challenges related to the privacy and ethical use of the documents.

- Different types of printing techniques, paper quality, ink types, scanning at different resolutions and degradation over time can affect the quality of the document. Because of these factors, it is very difficult for machine learning techniques also to detect the proper features and train the model accurately.
- Different types of feature extraction techniques and classification techniques have to be implemented because of the diversity in the document printed, handwritten, scanned, photocopy or digitally created.
- Machine learning and deep learning techniques need high computational environments to train the models properly.
- The use of sensitive data in document forensics raises questions about data privacy and regulatory compliance. If forensic analysis is not done properly, it can also be aggressive and have an impact on people's rights. The false-positive or false-negative results of a technique can lead to serious consequences. To prevent these kinds of mistakes, forensic models need to be precisely tested, particularly in situations with high risks. In order to train models on extremely specialized types of document fraud, provide artificial data which must be exactly the same copy of real document.
- In the future work, the models can be based on real-time detection algorithms with more speed up and accurate techniques. But the main challenge in the real-time scenario can be maintaining the accuracy and robustness with the low computational systems.
- With the help of deep learning methods, hybrid models also can be introduced which include the features of active and passive techniques. Active techniques like digital signatures or invisible patterns can be embedded along with passive techniques like ink density, texture, or noise patterns in the hybrid models.
- As current methods are struggling with the high computational data, future research may look at on-device processing methods that use less infrastructure, lightweight models, and algorithms with lower complexity.
- Automated fraud detection systems in the legal and banking sectors can use document forensics by using machine learning and deep learning methods to verify embedded data, such as watermarks, and evaluate inherent properties, including noise patterns. By enabling the real-time detection of manipulated or falsified papers, these solutions guarantee compliance and operational security.
- By lowering computational complexity without sacrificing accuracy, lightweight models like MobileNet and SqueezeNet may be used for document forensics in resource-constrained contexts. Performance is further optimized by different strategies like effective feature extraction, quantization, and model pruning. These models make it possible to identify forgeries in real time on devices with low computing power.
- Deep learning techniques like CNN can be implemented at the microlevel forgeries, such as minor pixel manipulations, font modifications, line-by-line tampering or transformer-based architectures for extremely detailed document analysis.

In this review paper, we have explored various passive techniques used to identify the source of the printer for digital documents. We discussed techniques based on different textures and patterns in documents and fonts, as well as the noise generated by printers. These feature extraction techniques aid in identifying printer brands and models. Different kinds of techniques used to identify the writer of the handwritten documents are also explored. Machine learning and deep learning techniques are discussed to extract the different features and for the purpose of classification. We have also pointed out the different kinds of challenges which may be encountered in the practical implementation of these techniques.

The study's findings indicate that passive procedures are preferable to active techniques since they can be applied directly to documents without requiring prior information. However, there are unresolved issues with passive forensics,

Significant opportunities exist for future document forensics research to enhance detection methods and increase flexibility in a variety of real-world scenarios.

such as achieving high accuracy, maintaining low feature dimensionality, robustness against added noise, low computing costs, the need for appropriate datasets, and effective classification methods. Although significant progress has been made with passive techniques, there is still potential for improvement because the methods proposed for feature selection, data extraction, data matching, and classification have limitations. For more reliable solutions, future research might concentrate on applying the latest technologies like block chain and deep learning. In order to make these technologies available to different industries, future initiatives should give priority to real-time detection systems and affordable tools because the ultimate goal of the document forgery detection is not to detect the forgeries but to prevent them and protect people and organizations completely from negative effects.

Author contributions Ranveer Sukhija performed original—writing and review protocol. Munish Kumar and M. K. Jindal presented supervision and final draft.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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